

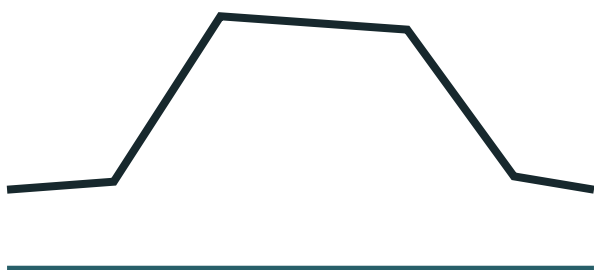
Vitreoretinal surgery and trauma

Parts 1 and 2 | The plane and the maintaining force

Name the mechanism before choosing an operation.

RRD

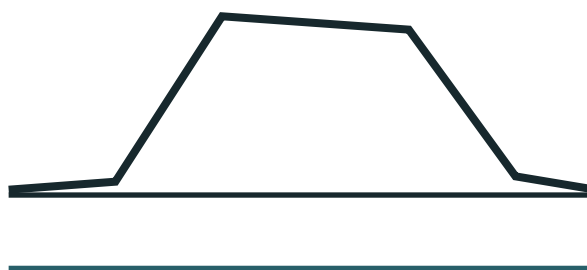
Subretinal separation



RPE

Retinoschisis

Split within retina



RPE

Original anatomical comparison: retina-RPE separation versus an intraretinal split.

In rhegmatogenous detachment, a break permits fluid beneath the neurosensory retina; repair closes the break and addresses relevant traction. Degenerative retinoschisis separates retinal layers and may remain stable without surgery. An outer-layer break alone is different from progressive detachment with communicating inner and outer breaks. Identify the actual plane before proposing repair.

Selection changes the trade-off

Lens, vitreous and break geometry influence the primary repair.

Finding	Implication
Young clear phakic eye	Consider lens-preserving external support
Suitable superior causative breaks	Assess pneumatic repair and positioning
Complex traction or PVR	Assess internal relief of the maintaining force
Counselling	Separate primary success, final attachment and function

A young phakic eye with suitable peripheral breaks may favour an external buckle to preserve the clear lens. Pneumatic repair requires an eligible break configuration and reliable positioning; its primary success and final attachment are different endpoints. Significant traction, complex breaks or PVR can favour vitrectomy. A complete break map changes the operation, not merely its dosage.

After surgery, localise the complication

The same pressure reading can result from different mechanisms.

Finding	Implication
Blocked aqueous route	Restore passage when anatomically indicated
Open angle with oil droplets	Balance outflow treatment and retinal support
Implant in the anterior chamber	Assess corneal and migration risk
Attached retina with poor acuity	Inspect the full optical and neural pathway

In an aphakic oil-filled eye, an obstructed inferior iridectomy can produce pupillary block: restoring the aqueous route targets the mechanism. With a patent opening and oil droplets in the angle, trabecular dysfunction is a competing cause. Oil removal may leave established pressure damage and must be balanced against the retina's continuing need for tamponade.

Retinitis: control activity and anticipate structural damage

Immune context and lesion anatomy shape urgency.

Finding	Implication
Peripheral necrosis with inflammation	Investigate relevant viral retinitis
Focal scar-adjacent active lesion	Consider toxoplasmic disease in context
Systemic immunosuppression	Broaden phenotype and organism assessment
After acute control	Monitor mechanical and macular sequelae

Peripheral confluent necrotising retinitis with substantial inflammation supports an acute retinal necrosis pathway; a focal active lesion beside a pigmented scar supports toxoplasmosis in the appropriate host. Aqueous or vitreous testing can refine uncertain infectious diagnoses. Even after organism control, necrotic retina remains vulnerable to breaks and detachment; steroid choice requires an antimicrobial strategy.

Trauma: do not end at the quiet wound

Late toxicity, macular change and neural damage need different decisions.

Finding	Implication
Possible retained ferrous fragment	Localise and assess toxic consequences
Recent small traumatic hole	Consider serial OCT in selected patients
Indirect optic-nerve injury	Balance uncertain benefit against systemic harm
New symptoms later	Reassess the structure suggested by the change

A quiet eye months after penetrating trauma can still contain a ferrous fragment causing progressive siderosis; localisation and toxic-function assessment remain important. A selected recent traumatic macular hole may close during observation. Indirect optic neuropathy with head injury presents a separate evidence problem: routine megadose steroids have no established visual benefit and systemic harm matters.